

NISHANK KUSHWAHA

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Department of Electrical Engineering (IDD), Indian Institute of Technology (BHU), Varanasi

EDUCATION

Degree / Certificate	Institute / Board	CPI / Percentage	Year
B.Tech–M.Tech (IDD), Electrical Engg.	Indian Institute of Technology (BHU), Varanasi	8.96 / 10	2024–Present
Senior Secondary (XII)	Modern Public School, Jhansi (CBSE)	88.6%	2024
Secondary (X)	Christ The King College, Jhansi (ICSE)	88.9%	2022

TECHNICAL SKILLS

Languages	C, C++, JavaScript (ES6+), Python
Web	HTML5, CSS3, Tailwind CSS, React.js, Next.js, Node.js, Express.js
Databases	MongoDB, SQL, Appwrite
Data Science	NumPy, Pandas, Matplotlib
DevOps & Tools	Git, GitHub, Docker, MATLAB/Simulink

PROJECTS

My_Git – Distributed Version Control System in C++	June 2026
- Built a Git-inspired distributed version control system from scratch in C++20, implementing content-addressable blob/tree/commit storage with a custom SHA-1 hashing engine and zlib-compressed object database. - Implemented branching, detached HEAD, merge-base discovery, three-way merge with conflict detection, LCS-based diff generation, repository integrity verification (fsck), and multi-lane ASCII commit DAG visualization. - Developed local and HTTP-based distributed workflows supporting push, fetch, pull, clone, along with a standalone HTTP server enabling remote repository synchronization over the network. Stack: C++20, STL, std::filesystem, zlib, cpp-httplib, CMake	
	GitHub

Real-Time Chat Application	October 2025
- Engineered a full-stack real-time messaging platform supporting private and group conversations using Socket.IO, enabling low-latency bidirectional communication and live message delivery. - Designed a secure authentication and account recovery system with JWT-based authorization, password hashing, OTP email verification, and password reset workflows using Nodemailer. - Built scalable messaging features including real-time notifications, unread message tracking, typing indicators, user search, and paginated message retrieval to improve performance for large conversations. Stack: React, Redux Toolkit, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS	
	GitHub — Live Demo

Full-Stack Image Gallery Application	August 2025
- Developed a cloud-based image management platform with secure authentication, allowing users to upload, organize, and manage personal image collections through a responsive web interface. - Integrated Cloudinary for scalable cloud storage and media delivery, supporting efficient image uploads, retrieval, and lifecycle management through RESTful APIs. - Implemented secure user authentication using JWT and OAuth-based login flows, along with state management and protected routes for personalized gallery access. Stack: React, Redux Toolkit, Node.js, Express, MongoDB, Cloudinary, Tailwind CSS	
	GitHub — Live Demo

ACHIEVEMENTS

- First Prize — HackOverflow Web Development Competition** at Prastuti'26, IIT (BHU) Varanasi.
- Attained **1200+ rating** on **Codeforces** through consistent competitive programming practice. [View Profile]

POSITIONS OF RESPONSIBILITY

- Technical Team Member** — Kashiyatra 2026 (Cultural Fest), IIT (BHU) Varanasi.
- Technical Team Member** — Prastuti 2026 (Technical Fest), IIT (BHU) Varanasi.

AREAS OF INTEREST

Full-Stack Web Development	Competitive Programming	Data Structures & Algorithms
Data Analysis with Python	Open Source Contribution	System Design

HOBBIES & OTHER INTERESTS

Problem Solving · Building Web Applications · Exploring AI Technologies · Data Visualization